Week 0 (CS 418 @ UIC)

Week 0 Slide Deck

An Experiment With Quarto

Jack Bandy 2026

This is a demo slide deck for reference and experimentation.

Harold Washington Library

CS 418 · Week 0 · 🟠 Harold Washington Library 🟠


Harold Washington Library
State/Van Buren
1W
400S
Elevator
Danger - Keep Off Tracks
High Voltage
Danger
No clearance for person between trains
Let's take this show on the road.
xfinity
America's
wake u
Missing your medic
worse than missing y

Content Slide

Testing content slide.

- Lorem ipsum dolor sit amet
- Consectetur adipiscing elit
- Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua
- Test quote

> *“The greatest value of a picture is when it forces us to notice what we never expected to see.” — John Tukey*

Incremental Bullets

- display this first
- display this second
- display this third
- display this fourth
- display this fifth

Tabsets

Drop a finger on a random spot of the spinning globe: **Water** or **Land**? About 75% of Earth's surface is water, so each toss is a coin flip weighted by `p`.

Python R Julia JavaScript

```python
import random

def globe_toss(p=0.75):
    return "Water" if random.random() < p else "Land"
```

Pretty Code

- Syntax highlighting works in revealjs slides
- Code uses the theme's Courier stack

```python
import polars as pl


def summarize_ridership(stations: pl.DataFrame) -> pl.DataFrame:
    return (
        stations
        .group_by("line")
        .len(name="station_count")
        .sort("station_count", descending=True)
    )
```

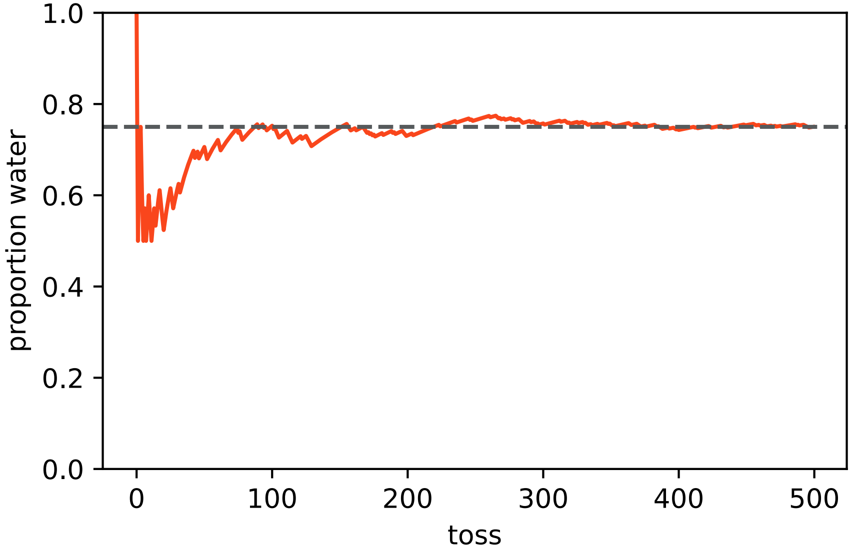
Line Highlighting

```python
import numpy as np
import polars as pl

stations = pl.read_csv("cta_stations.csv")
orange = stations.filter(pl.col("line") == "Orange")

by_access = orange.group_by("accessible").len(name="station_count")
share = np.round(
    by_access["station_count"] / by_access["station_count"].sum(),
    2,
)

print(by_access.with_columns(accessible_share=share))
```

Code + Figure

The globe toss from earlier: the proportion of water converges on `p`.

```
import numpy as np
import matplotlib.pyplot as plt
water = np.random.default_rng(418).random(500) < 0.75
running = np.cumsum(water) / np.arange(1, 501)
fig, ax = plt.subplots()
ax.plot(running, color="#f9461c")
ax.axhline(0.75, color="#565a5c", linestyle="--")
ax.set(xlabel="toss", ylabel="proportion water", ylim=(0, 1))
```



Column Layout

Course Pattern

- Ask a data question
- Find the relevant table
- Check the units
- Make a readable view

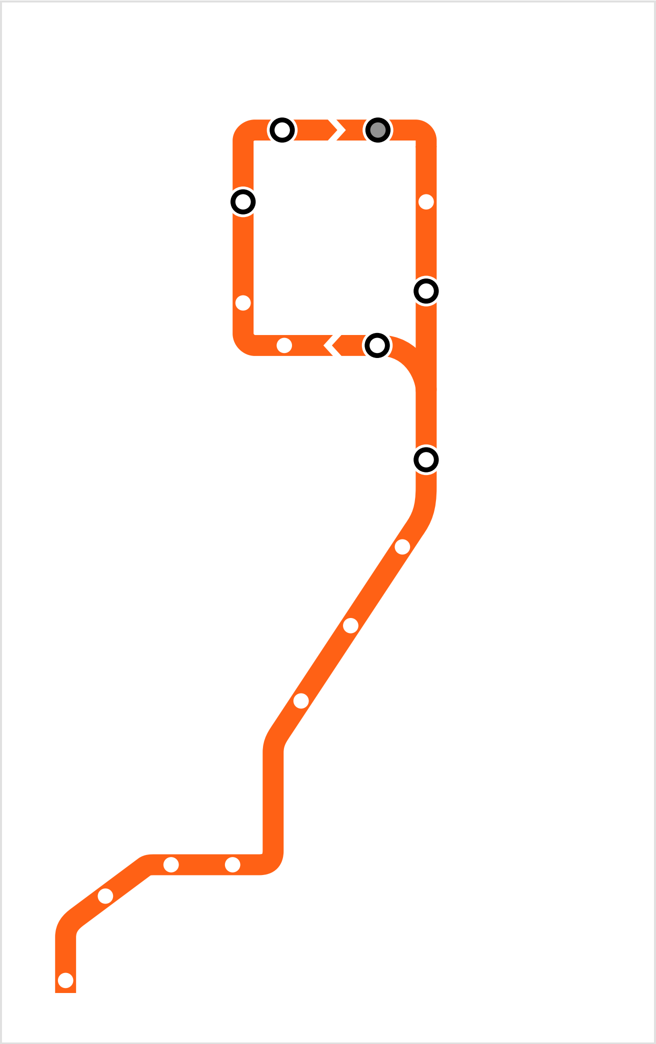
Week	Focus
0	Setup and orientation
1	Blah blah
2	Blah blah blah
3	Blah blah blah blah

Auto-Animate

Auto-Animate

Image & Embed Slides

Image Frame Slide

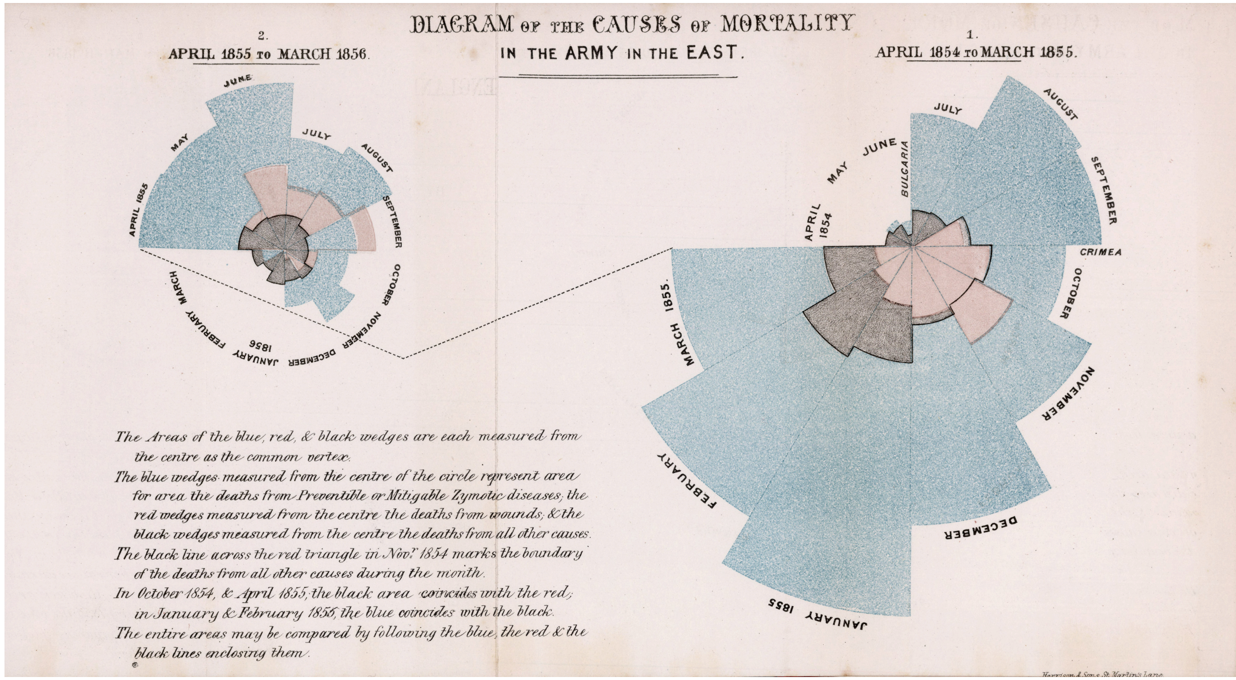


Florence Nightingale’s Mortality Diagram (1858)

A **polar area chart** showing causes of mortality in the Crimean War:

- Blue: deaths from preventable disease
- Red: deaths from wounds
- Black: other causes

Used to advocate for sanitary reforms in military hospitals.



Florence Nightingale’s Mortality Diagram (1858)

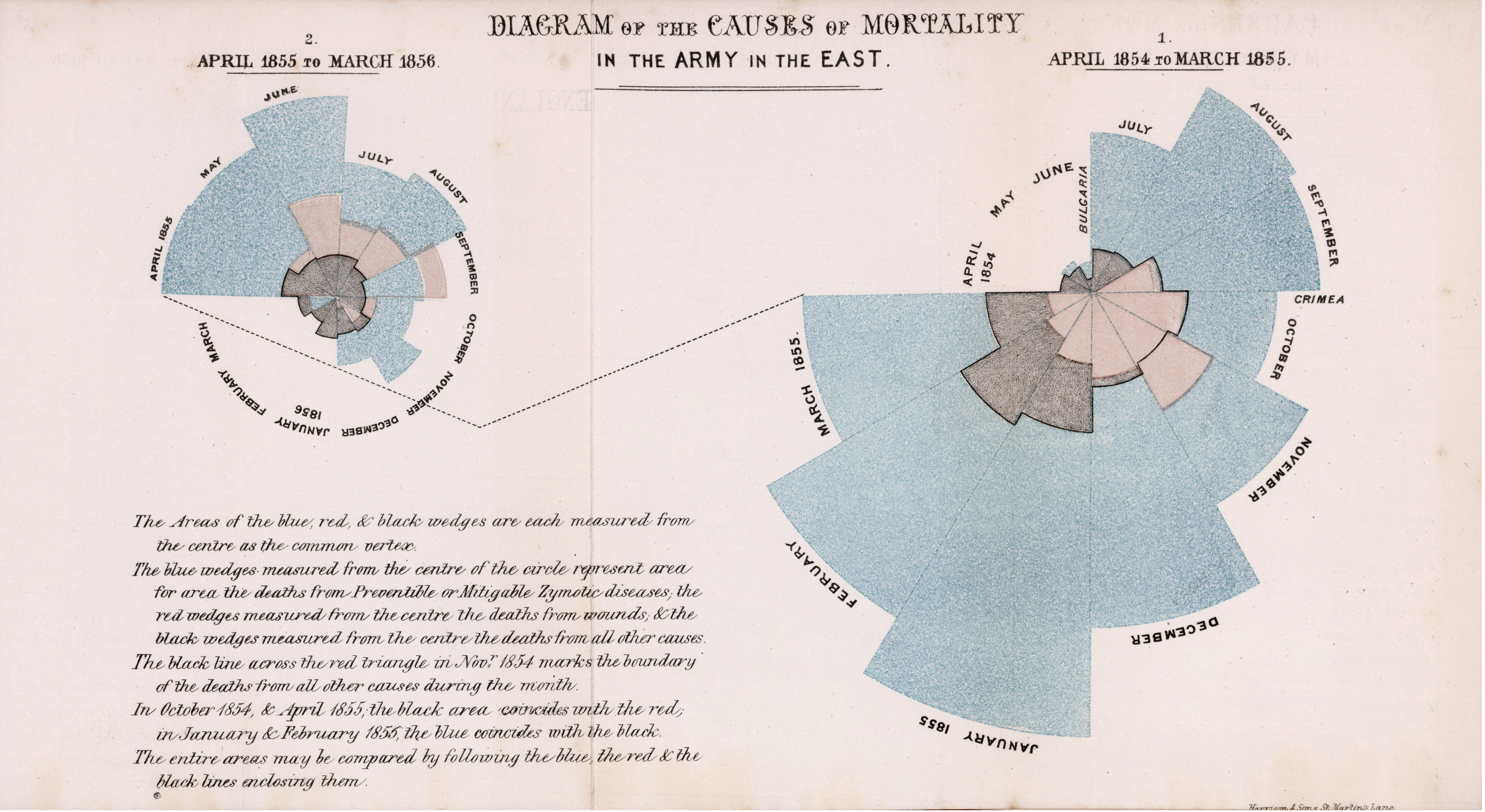
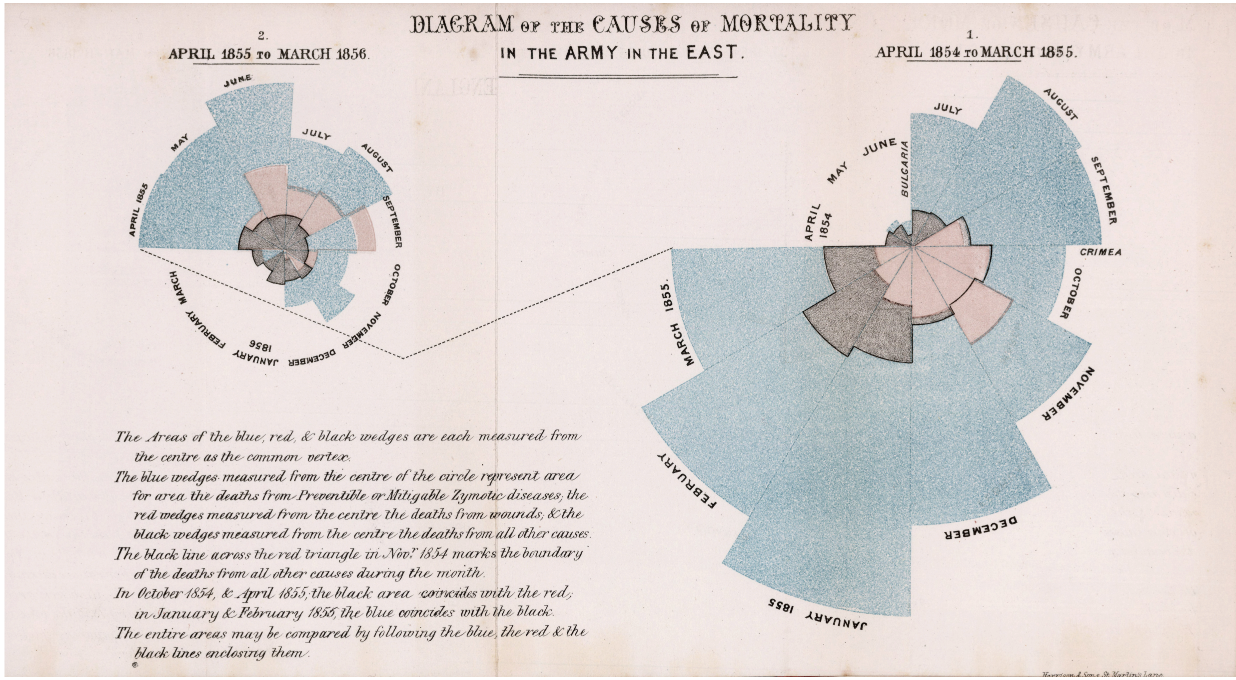
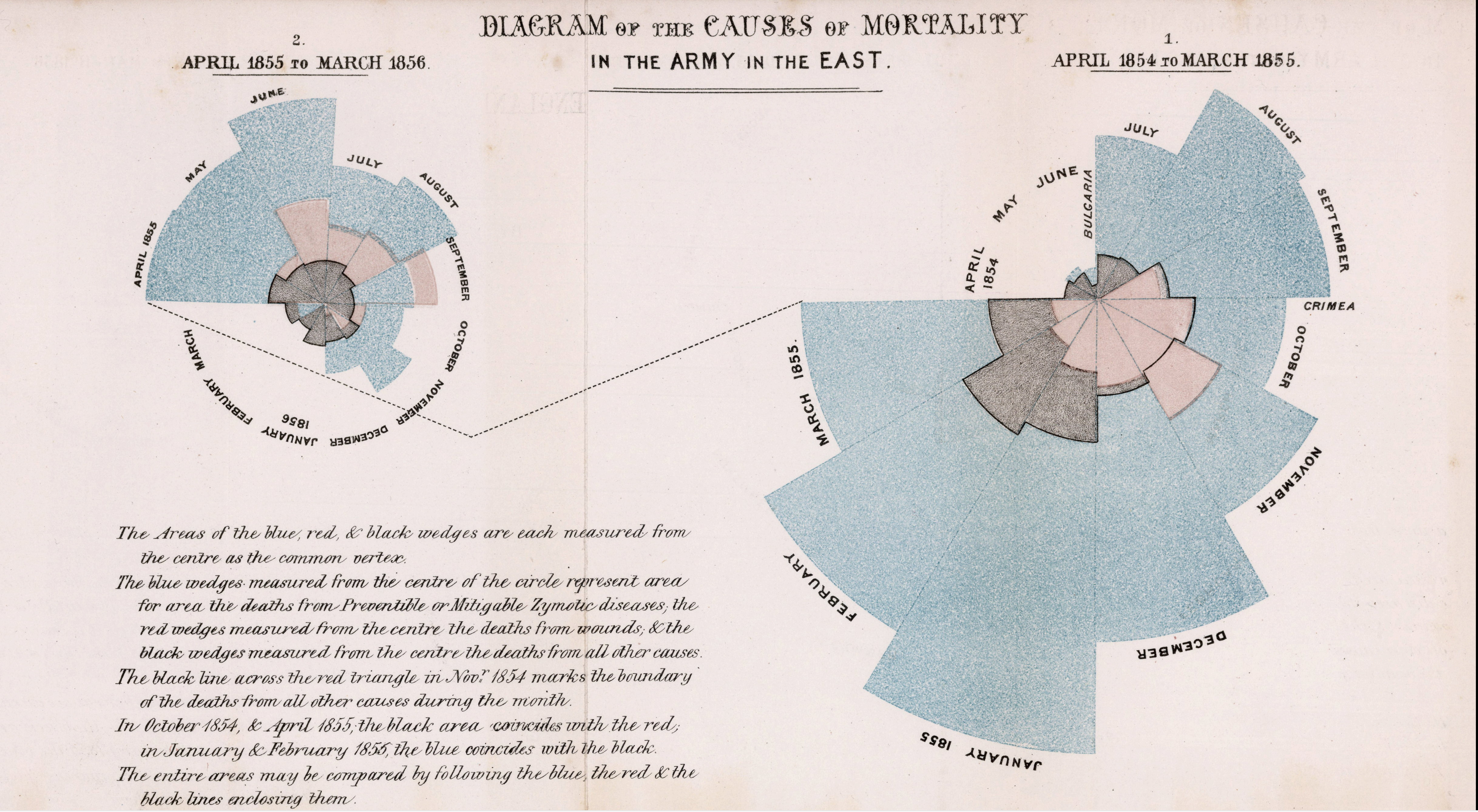


DIAGRAM OF THE CAUSES OF MORTALITY
IN THE ARMY IN THE EAST.
2.
APRIL 1855 TO MARCH 1856.
1.
APRIL 1854 TO MARCH 1855.
JUNE
JULY
AUGUST
SEPTEMBER
OCTOBER
NOVEMBER
DECEMBER
JANUARY 1856
FEBRUARY
MARCH
APRIL 1855
MAY
APRIL 1854
MAY
JUNE
BULGARIA
JULY
AUGUST
SEPTEMBER
CRIMEA
OCTOBER
NOVEMBER
DECEMBER
JANUARY 1855
FEBRUARY
MARCH 1855.
The Areas of the blue, red, & black wedges are each measured from the centre as the common vertex.
The blue wedges measured from the centre of the circle represent area for area the deaths from Preventible or Mitigable Zymotic diseases; the red wedges measured from the centre the deaths from wounds; & the black wedges measured from the centre the deaths from all other causes.
The black line across the red triangle in Nov.r 1854 marks the boundary of the deaths from all other causes during the month.
In October 1854, & April 1855, the black area coincides with the red; in January & February 1856, the blue coincides with the black.
The entire areas may be compared by following the blue, the red & the black lines enclosing them.

follow-up slide for nightingale

- Lorem ipsum dolor sit amet,
- consectetur adipiscing elit
- Sed do eiusmod tempor incididunt
- ut labore et dolore magna

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University of Illinois Chicago

embedded web page: https://www.uic.edu

Sources

1. Nightingale, Florence. "Diagram of the causes of mortality in the army in the East." *Notes on Matters Affecting the Health, Efficiency, and Hospital Administration of the British Army*, 1858. Image via Wikimedia Commons: https://commons.wikimedia.org/wiki/File:Nightingale-mortality.jpg. Public domain.
2. Slides developed using materials from Elena Zheleva and Gonzalo Bello Lander, the Berkeley DS 100 team, Marine Carpuat, and Brian Ziebart.
3. Slide deck built with Quarto revealjs.
4. Last modified and compiled June 23, 22:12h Central (Chicago) time.
5. Title font is Big Shoulders; Body font is Libre Franklin.